

MATSUE, Japan

WMO: 477410

Lat: 35.4572N

Lon: 133.0656E

Elev: 73

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.0	31.6	20.4	15.3	37.1	22.6	17.1	38.0	30.8	35.5	28.4	36.4	8.4	270	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	92.3	78.4	90.1	78.0	88.0	77.3	79.9	88.4	79.0	87.2	78.1	85.8	10.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.6	143.8	84.2	76.7	139.3	83.5	75.8	135.4	82.9	43.5	88.6	42.6	87.5	41.7	86.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.3	21.9	19.0	DB	26.1	96.0	2.3	1.9	24.4	97.3	23.0	98.4	21.7	99.4	20.0	100.8
			WB	24.7	81.5	2.4	0.9	23.0	82.2	21.6	82.7	20.2	83.2	18.5	83.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	40.3	41.2	46.7	55.9	64.9	71.7	79.3	81.8	73.8	63.9	53.9	44.8	(6)
(7)		DBStd	15.00	4.20	4.79	5.60	5.81	4.62	4.10	4.57	4.00	4.91	4.84	5.24	4.90	(7)
(8)		HDD50	893	302	250	135	14	0	0	0	0	0	0	20	172	(8)
(9)		HDD65	3374	766	666	567	276	58	1	0	1	78	334	627	(9)	
(10)		CDD50	4530	1	4	32	192	463	650	909	987	714	431	137	10	(10)
(11)		CDD65	1538	0	0	0	4	57	202	444	522	265	43	1	0	(11)
(12)		CDH74	12429	0	0	1	30	302	933	4033	5445	1561	123	1	0	(12)
(13)		CDH80	4156	0	0	0	1	33	169	1413	2144	381	15	0	0	(13)
(14)	Wind	WSAvg	7.4	8.4	7.7	8.0	8.1	7.7	7.0	7.5	6.9	6.4	6.1	6.7	8.2	(14)
(15)	Precipitation	PrecAvg	69.8	6.4	4.6	5.1	4.3	5.5	6.5	9.1	5.2	8.3	5.2	5.1	6.1	(15)
(16)		PrecMax	87.6	9.6	7.9	8.4	8.8	14.8	14.7	23.5	11.6	19.0	14.1	8.9	10.6	(16)
(17)		PrecMin	58.1	3.8	2.8	2.4	1.5	1.2	1.0	0.3	0.9	2.0	1.9	1.4	1.5	(17)
(18)		PrecStd	8.1	1.6	1.3	1.6	1.8	3.3	3.7	6.1	3.0	4.6	3.3	2.0	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	63.4	69.6	77.8	83.7	87.5	93.5	95.9	91.0	81.6	72.0	61.7	(19)
(20)			MCWB	50.9	54.3	56.0	61.9	66.6	73.2	78.6	78.6	77.5	71.5	61.8	53.7	(20)
(21)		2%	DB	52.4	56.5	64.2	73.4	80.2	84.2	91.2	93.3	87.3	77.2	68.1	57.8	(21)
(22)			MCWB	45.9	49.4	53.6	60.3	64.6	71.6	78.4	78.4	76.1	68.1	59.8	51.3	(22)
(23)		5%	DB	49.7	52.8	60.5	70.0	77.4	81.7	89.1	91.3	84.3	74.3	65.5	55.3	(23)
(24)			MCWB	44.3	46.6	51.7	58.2	63.6	71.0	77.9	78.4	74.7	65.8	58.6	49.5	(24)
(25)		10%	DB	47.4	49.8	57.0	66.7	74.7	79.3	87.2	89.2	81.5	72.0	62.9	52.8	(25)
(26)			MCWB	42.5	44.4	49.9	56.7	62.4	70.2	77.2	78.0	73.4	64.2	57.0	47.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	56.2	58.6	66.5	70.6	76.8	80.7	81.5	79.2	73.1	65.0	56.7	(27)
(28)			MCDB	56.2	62.2	65.9	72.9	76.0	83.8	89.4	90.5	87.0	79.3	68.5	58.5	(28)
(29)		2%	WB	47.4	51.1	55.5	62.5	68.1	75.0	79.6	80.2	77.4	70.7	62.3	52.9	(29)
(30)			MCDB	50.7	55.0	61.0	70.2	73.8	80.4	87.9	89.0	84.5	75.2	65.8	56.3	(30)
(31)		5%	WB	44.9	47.7	53.0	60.2	66.3	73.5	78.7	79.4	76.2	68.2	60.1	50.4	(31)
(32)			MCDB	48.5	51.7	58.4	67.2	73.7	78.9	86.8	87.8	82.4	72.1	64.2	53.9	(32)
(33)		10%	WB	43.0	44.9	50.8	58.3	64.6	71.8	77.8	78.6	74.8	65.7	57.7	48.1	(33)
(34)			MCDB	46.8	48.8	56.4	64.3	71.8	76.9	85.3	86.7	80.1	70.1	62.0	52.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	12.7	15.4	16.8	16.0	12.8	11.4	12.6	12.1	13.8	13.6	11.6	(35)
(36)			MCDBR	15.7	19.8	21.8	22.1	21.2	16.7	14.7	15.5	15.0	16.4	17.5	16.5	(36)
(37)			MCWBR	10.5	13.2	13.3	11.2	9.5	7.2	5.9	5.6	6.4	8.3	10.5	10.9	(37)
(38)		5% WB	MCDBR	14.6	17.3	18.6	17.8	16.3	12.2	13.1	13.7	12.9	12.6	14.7	14.9	(38)
(39)			MCWBR	10.9	13.0	12.8	10.4	8.9	6.3	5.8	5.6	6.0	7.5	10.1	11.1	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.436	0.471	0.539	0.557	0.554	0.572	0.569	0.531	0.490	0.451	0.434	0.428	(40)
(41)	taud		2.003	1.944	1.809	1.820	1.885	1.920	1.987	2.097	2.162	2.157	2.110	2.048	(41)	
(42)	Ebn at Noon		228	237	232	237	239	235	234	241	245	242	229	222	(42)	
(43)	Edh at Noon		43	51	64	66	63	61	56	50	44	41	39	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	548	783	1152	1515	1675	1567	1556	1615	1250	1020	696	497	(44)	
(45)		RadStd	43	77	102	124	162	121	230	176	107	90	56	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page